

Executive KPI



Here is the structured formulation for your Executive Dashboard KPIs. The formulas have been standardized mathematically to aggregate individual spreadsheet columns into clean, actionable, percentage-based or count-based metrics suitable for a dashboard backend or spreadsheet implementation.

1. Quality & Accuracy Metrics

ACE Trust Score

Calculates the overall scientific and operational reliability of answers delivered to users.

Formula:

$$\text{ACE Trust Score} = 0.40(A_{\text{sci}}) + 0.20(A_{\text{dom}}) + 0.10(S_{\text{lnk}}) + 0.10(E_{\text{exp}}) + 0.1$$

Sub-Component Breakdowns:

Scientific Accuracy (A_{sci}): % of "Yes" in Answer Scientifically Correct?

Domain Accuracy (A_{dom}): Average pass rate of domain questions:

$$\frac{\% \text{ Pass (Weather Q)} + \% \text{ Pass (Mandi Price Q)} + \% \text{ Pass (Scheme Q)}}{3}$$

Source Links (S_{lnk}): % of "Yes" in Source Links Provided?

Expert Accuracy (E_{exp}): % of "Yes" in both Correct Expert Name? AND Expert Name Displayed?

Translation Quality (Q_{trn}): Average score or passing rate percentage of Translation Quality .

Cross Channel Consistency (C_{chn}): % of "Yes" in WhatsApp vs Web Answer Match?

Farmer Experience Score

Measures the operational efficiency and end-user usability of the communication channel.

Formula:

$$\text{Farmer Experience Score} = 0.30(S_{\text{rsp}}) + 0.20(S_{\text{sla}}) + 0.20(V_{\text{io}}) + 0.15(Q_{\text{trn}})$$

Sub-Component Breakdowns:

Response Speed (S_{rsp}): Derived from `Response Time (mins)` . Scaled mapping value (e.g., 100% if ≤ 15 mins, scaling down to 0% if > 120 mins).

SLA Status (S_{sla}): % of "Met" or "Pass" in `SLA Status` .

Voice Capability (V_{io}): Composite pass rate of voice mechanics:

$$\frac{\% \text{ Yes (Voice Input Working)} + \% \text{ Yes (Voice Output Working)} + \text{Av}}{4}$$

Translation Quality (Q_{trn}): Same percentage metric as used in the Trust Score.

Notification Experience (N_{exp}): % of entries where `Notification Received? = "Yes"` AND `Notification on Same Thread? = "Yes"` AND `Notification Linked Correct Q-ID? = "Yes"` .

2. Risk & Operational Health Metrics

Critical Failures Today

A direct automated volume count tracking systemic or data integrity breaking bugs over a 24-hour cycle.

Formula Logic:

$$\text{Critical Failures Today} = \sum (\text{Critical Scientific Failures} + \text{Data Integrity})$$

Operational Definition: Count every row evaluated today where:

`Answer Scientifically Correct? = "No"` OR

`Weather / Mandi / Scheme Q Correct? = "No"` (when applicable) OR

`Question Saved in DB? = "No"` OR

`Answer Saved in DB? = "No"` OR

`Q-ID Consistent Across Systems? = "No"` OR

Defect Severity = "Critical" / "Blocker"

Release Health

Evaluates the functional stability of the current active environment/build.

Formula Logic:

$$\text{Release Health \%} = \left(\frac{\text{Total Passed Tests}}{\text{Total Tests Executed}} \times 100 \right) - \text{Count}(\text{Critical Defects})$$

Operational Definition:

Pass Rate: Derived directly from the Overall Test Status column metrics.

Critical Defects: Rows where Defect Severity = "Critical" .

Data Integrity Failures: Rows where Question Saved in DB? , Answer Saved in DB? , or Q-ID Consistent Across Systems? evaluate to "No" .

3. Optimization & Diagnostics Metrics

Biggest Bottleneck

Identifies the exact phase slowing down content delivery or review cycles.

Formula Logic:

$$\text{Biggest Bottleneck} = \text{Target Stage } X \text{ where } \overline{\text{TAT}}_X = \max(\overline{\text{Author TAT}}, \overline{\text{Editor TAT}}, \overline{\text{Reviewer TAT}}, \overline{\text{QA TAT}})$$

Operational Definition: Run an average calculation (AVERAGE()) across each individual Turnaround Time (TAT) column. Output the string name of the column header that holds the highest value.

Weakest Module

Identifies the core functional domain underperforming across accuracy parameters.

Formula Logic:

$$\text{Weakest Module} = \text{Module Name of } \min(A_{\text{Weather}}, A_{\text{Mandi}}, A_{\text{Scheme}}, Q_{\text{Translati}})$$

Operational Definition: Calculate individual success rates for each pillar. Return the system name containing the lowest calculated percentage metric.

4. Analytical Tracking

Trend

Monitors longitudinal velocity changes across engineering iterations.

Formula Logic:

$$\Delta_{Trend} = Metric_{Current\ Build/Sprint} - Metric_{Previous\ Build/Sprint}$$

Dashboard Visual Mapping:

Trust / Farmer Experience / Response Time Trends: $\Delta > 0$ is a positive indicator (Green ↑).

Defect / Review TAT Trends: $\Delta < 0$ is a positive indicator (Green ↓).